Théo Matricon

2025–current

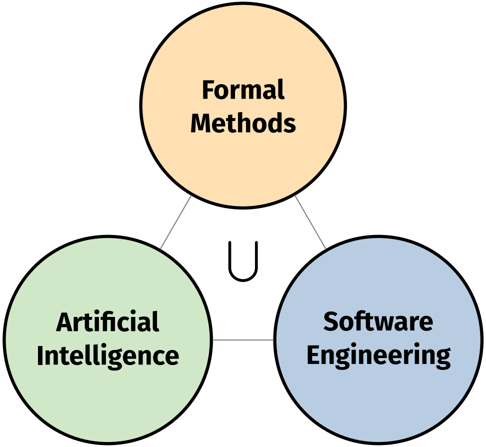
Postdoc with Mathieu Acher
Code Generation, Reproducibility
DiverSE, Software Engineering team
IRISA, Rennes

2021–2024

PhD supervised by Nathanaël Fijalkow
Scaling domain agnostic techniques for program synthesis
M2F, Formal Methods team
LaBRI, Bordeaux

Medical Condition (RQTH)

Research Contributions



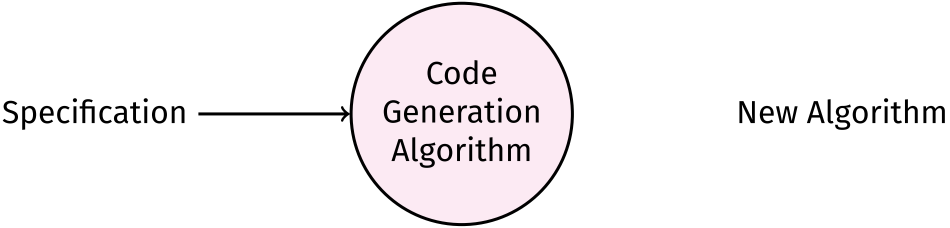
Selected Publications

- 2026, 2 TOSEM ($Q1$) accepts with major revisions
- 2026, MSR (A)
- 2025, TACAS (A)
- 2025, AAAI (top 20%) (A^{*})
- 2025, IST ($Q1$)

Postdoc

- 2022, AAAI (top 20%) (A^{*})

PhD Motivation



PhD Motivation: In Practice

Specifications

Logic

$$\forall a, b$$
$$f(a,b) \geq a$$
$$f(a,b) \geq b$$
$$f(a,b) \in \{a,b\}$$

Examples

$$f(1,5) = 5$$
$$f(2,1) = 2$$
$$f(-3,-9) = -3$$

Natural Language

‘Write a function that takes the maximum of its two arguments.’

Produced Algorithm

```
def max(a: int, b: int) ->int:
    if a <=b:
        return b
    else:
        return a
```

Program Synthesis: Problem

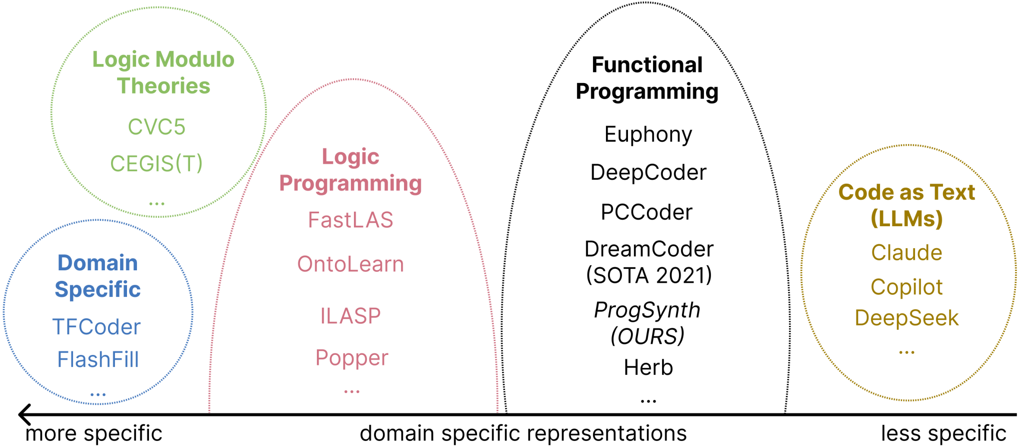
Input

- the search space G:
  a deterministic tree grammar
- a specification $\mathcal{C}$:
  checks if a program $p \in \mathcal{L}(G)$ matches the specification

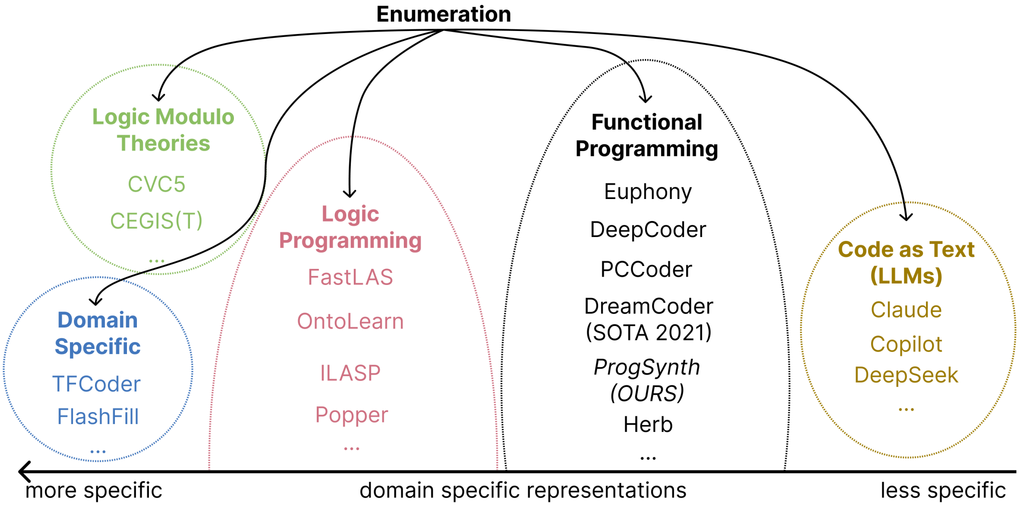
Output

- a program in the search space that matches the specification:
  $p \in \mathcal{L}(G)$ such that $\mathcal{C}(p) = \checkmark$

Program Synthesis: Frameworks



Program Synthesis: Frameworks



Program Synthesis: Enumeration Problem

Input

- the search space G:
  a deterministic tree grammar with a cost for each tree
- a specification $\mathcal{C}$:
  checks if a program $p \in \mathcal{L}(G)$ matches the specification

Goal

- enumeration of programs in order of non-decreasing costs

Delay

- time complexity between enumeration of two successive programs
  in terms of number of enumerated programs

Enumeration Algorithm Before Our Contribution

priority queue: S pairs of (program, cost)

$(p_1, c_1) \quad (p_2, c_2) \quad \cdots \quad (p_S, c_S)$

pop

p_n

push k successors

check if solution

$\mathcal{C}(p_n) = \checkmark$

At step n

$S = O(n)$

$k = O(1)$

Our Contribution: Constant Delay

bucket queue: M buckets of (program**s**, cost)

$(\mathcal{P}_1, c_1)$ $(\mathcal{P}_2, c_2)$ $\cdots$ $(\mathcal{P}_M, c_M)$

pop

$\mathcal{P}$

for each $p_n \in \mathcal{P}$

p_n

push k successors

check if solution

$\mathcal{C}(p_n) = \checkmark$

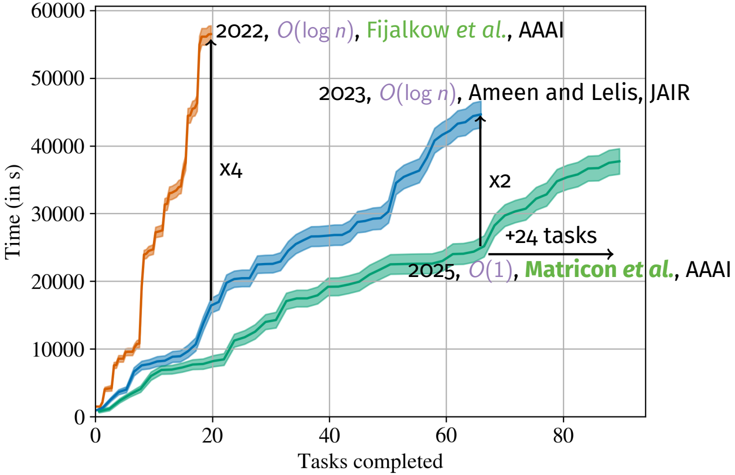
At step n

$M = O(1)$
$k = O(1)$

Our Contribution: Empirical Results

Major Papers

- 2017, ML + Enum., Balog *et al.*, ICLR
- 2018, $O(\log n)$, Lee *et al.*, PLDI



Our Contribution: Summary

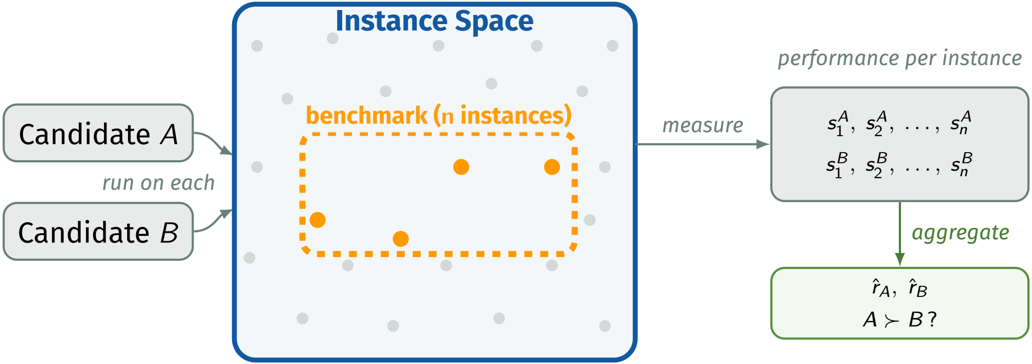
We prove bounded differences in cost:
$\exists M, \forall n, \ cost\_next(p_n) - cost(p_n) \leq M$

This implies: priority queues $O(\log n) \rightarrow$ bucket queues $O(1)$

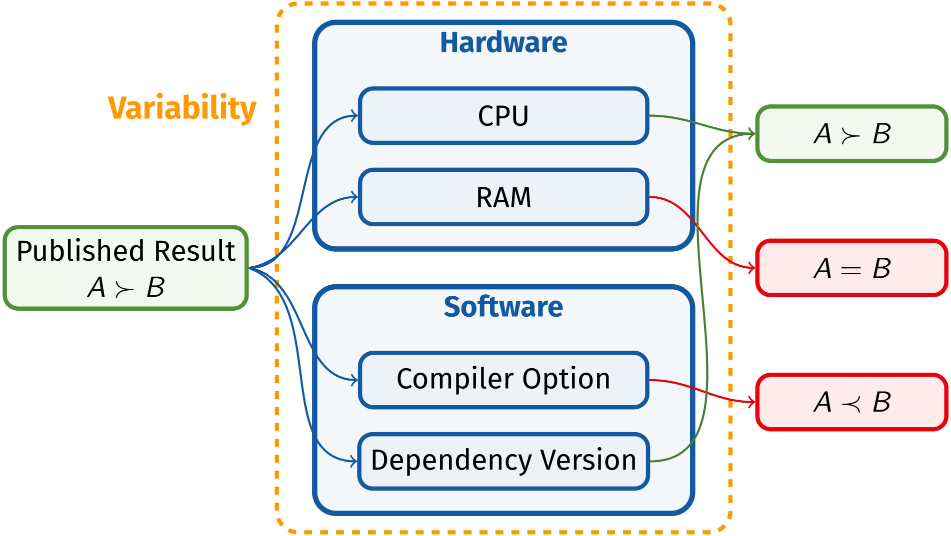
Impact

Published in **AAAI 2025** (+oral: 20% of accepted papers).
Fastest ranked enumeration for program synthesis **in practice**.
First algorithm with $O(1)$ delay $\rightarrow$ closes open question.

Evaluations



Evaluations: In Practice



Duplyakin *et al.*, “In datacenter performance, the only constant is change”, CCGRID, IEEE, 2020

My Research Project

Observations

- Exponential growth in scientific publications **[1]**
- Reproducibility crisis in computational science
- Evaluations are a key aspect of reproducibility

[1] Bornmann and Mutz, "Growth rates of modern science: A bibliometric analysis based on the number of publications and cited references", JAIST

My Research Project

Observations

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How to design reliable and efficient evaluations?

Reliable $\rightarrow$ formal guarantees

Efficient $\rightarrow$ reduce cost of evaluations

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Main Challenges

SE Modelling the huge hardware and software variability

AI Predicting heterogeneous evaluation data

FM Giving formal guarantees while still being efficient

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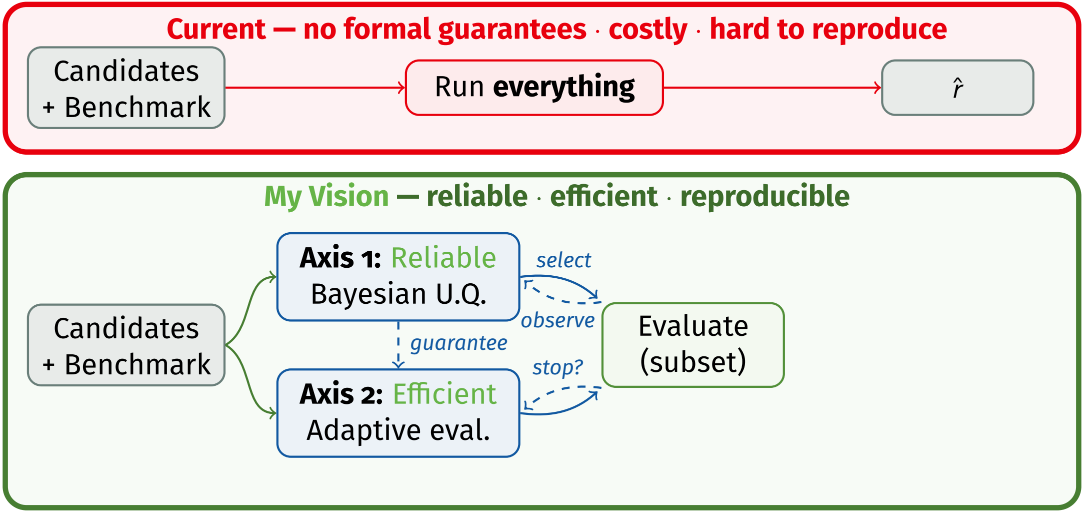
My Research Project

Current — no formal guarantees · costly · hard to reproduce

Candidates + Benchmark → Run **everything** → $\hat{r}$

My Research Project

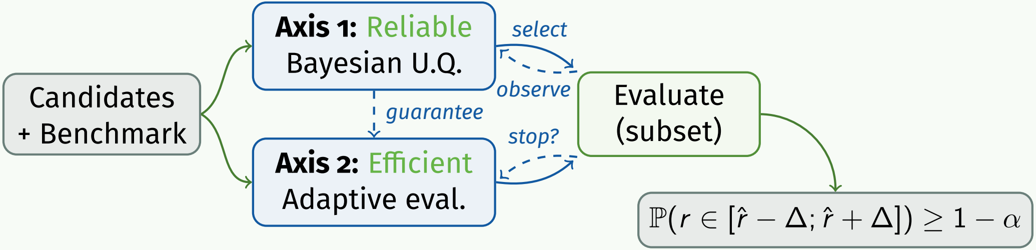
My Research Project



My Research Project

Current — no formal guarantees · costly · hard to reproduce

My Vision — reliable · efficient · reproducible



Classic Approach vs. Our Approach

Classic Approach

What is evaluated

Run all n instances — exhaustively

Our Approach

What is evaluated

Adaptive subset + early stopping
($\sim$50% fewer runs **[1]**)

[1] Matricon *et al.*, "Efficiently ranking software variants with tiny benchmarks" under revision at TOSEM

Classic Approach vs. Our Approach

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What is evaluated

Run all *n* instances — exhaustively

Statistical framework

assumes i.i.d. & normality

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Adaptive subset + early stopping (∼50% fewer runs **[1]**)

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Adjustable complexity

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Statistical framework

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Variability

Single environment assumed
results may not transfer

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What is evaluated

Adaptive subset + early stopping
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Statistical framework

Adjustable complexity

Variability

Multi-environment hierarchical model
quantifies generalizability

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Classic Approach vs. Our Approach

Classic Approach	Our Approach
What is evaluated	**What is evaluated**
Run all n instances — exhaustively	Adaptive subset + early stopping ($\sim$50% fewer runs **[1]**)
Statistical framework	**Statistical framework**
assumes i.i.d. & normality	Adjustable complexity
Variability	**Variability**
Single environment assumed results may not transfer	Multi-environment hierarchical model quantifies generalizability
Reproducibility	**Reproducibility**
binary: reproduced or not	graded: exact $\rightarrow$ ordinal $\rightarrow$ trend

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Axis 1: Reliable Evaluations

Variability

CPU-A • GCC-12 • Linux

$\hat{r}_A = 0.82 \quad \hat{r}_B = 0.75 \quad A \succ B$

CPU-B • GCC-14 • Linux

$\hat{r}_A = 0.79 \quad \hat{r}_B = 0.77 \quad A \succ B$

CPU-C • Clang • macOS

$\hat{r}_A = 0.71 \quad \hat{r}_B = 0.74 \quad A \prec B$

inconsistent?

Ernst, "Bayesian hierarchical modelling for tailoring metric thresholds", MSR, 2018

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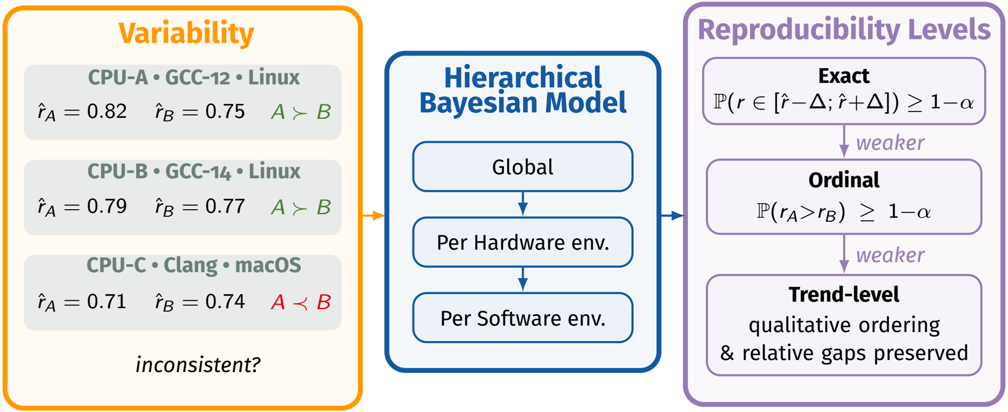
inconsistent?

Hierarchical Bayesian Model

Global → Per Hardware env. → Per Software env.

Ernst, "Bayesian hierarchical modelling for tailoring metric thresholds", MSR, 2018

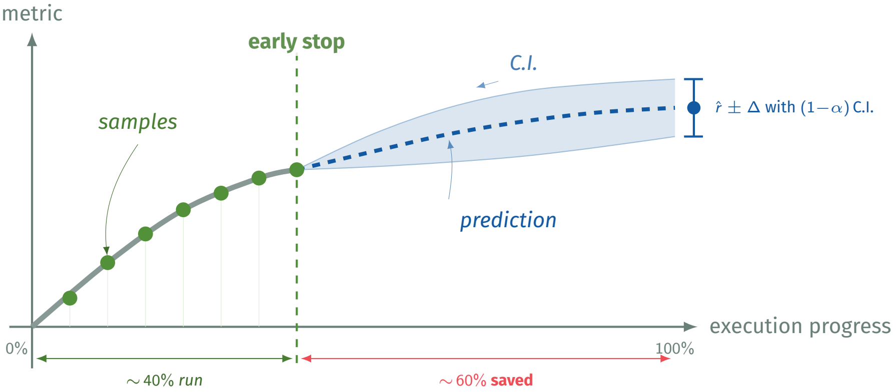
Axis 1: Reliable Evaluations



Ernst, “Bayesian hierarchical modelling for tailoring metric thresholds”, MSR, 2018

Axis 2: Efficient Evaluations

Execution on an instance is not atomic



Matricon *et al.*, "Statistical Comparison of Algorithm Performance Through Instance Selection" CP 2021

Integration

Transversal project with many applications in empirical fields

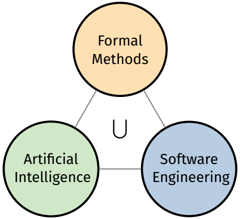
→ CRIStAL, UMR 9189, Lille, **Spirals** (SEEDS)

Frugality and Evaluations: Clément Quinton, Romain Rouvoy
Trust: Pierre Bourhis (Citadelle)
Statistics: Scool team

→ IRISA, UMR 6074, Rennes, **DiverSE** (rRaise)

Reproducibility: Mathieu Acher, Olivier Barais, Djamel E. Khelladi
Frugality: *Romain Lefeuvre*, Quentin Perez

Théo Matricon

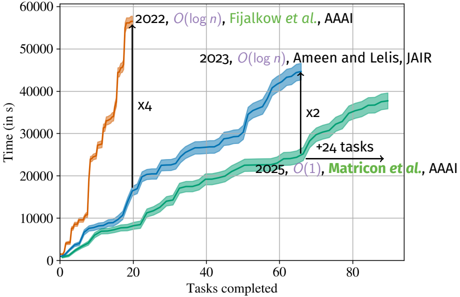


Selected Publications

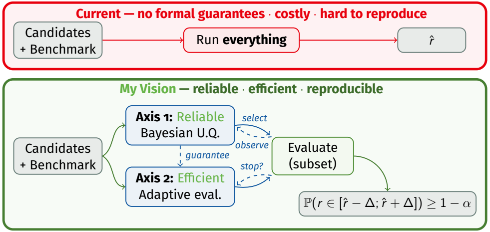
2 AAAI →

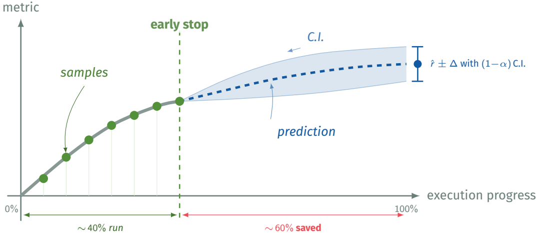
IST, MSR, 2 TOSEM

TACAS



Research Project





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